Other

SWAT+ Documentation

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SWAT+ can change model parameters from inside the model, using two kinds of calibration data.

Hard data are measured time series at a point, such as daily streamflow at a gauge, sediment or nutrient loads, or groundwater levels. Fitting the model to hard data is done by an external tool (SWAT+ Toolbox, SWAT-CUP, a script) that writes a parameter change list to `calibration.cal`, runs SWAT+, compares the output with the observations and repeats. SWAT+ applies the change list at start-up, before the simulation begins.

Soft data are estimates of individual processes that are not measured at the site, or only as long-term averages: the share of precipitation that leaves as surface runoff or evapotranspiration, the baseflow share of streamflow, or an average county crop yield (Seibert and McDonnell, 2002; Arnold et al., 2015). SWAT+ fits these itself. After the normal run it adjusts a few water balance and plant parameters one at a time, rerunning the simulation after each change, until the average annual values for each land-use group match the targets.

The usual order is soft calibration first, so that each part of the water balance is about right, then hard calibration against the gauge record, which should then need mostly adjustments to peaks and recessions.

Pages

- Hard calibration. How SWAT+ applies a parameter change list, change types and object types.
- Parameter change files. Layout of `cal_parms.cal` and `calibration.cal`, and every parameter name SWAT+ accepts.
- Conditions and regionalization. Applying a change only to objects that match a condition, such as a soil group, land use or calibration group.
- Soft calibration. The `codes.sft` switches, the targets in each `.sft` file, the adjustment steps and the output files.
- Sensitivity workflow. Running a sensitivity analysis with an external sampler that writes `calibration.cal`.

References

Arnold, J. G., Youssef, M. A., Yen, H., White, M. J., Sheshukov, A. Y., Sadeghi, A. M., Moriasi, D. N., Steiner, J. L., Amatya, D. M., Skaggs, R. W., Haney, E. B., Jeong, J., Arabi, M., and Gowda, P. H. (2015). Hydrological processes and model representation: impact of soft data on calibration. Transactions of the ASABE, 58(6), 1637–1660.

Seibert, J., and McDonnell, J. J. (2002). On the dialog between experimentalist and modeler in catchment hydrology: use of soft data for multicriteria model calibration. Water Resources Research, 38(11), 1241.

Sources

- Calibration start-up: `src/proc_cal.f90`, called from `src/main.f90.in` before the simulation.
- Hard calibration changes: `src/cal_conditions.f90` and `src/cal_parm_select.f90`.
- Soft calibration: `src/calsoft_control.f90`, run from `main.f90.in` after the base simulation when `cal_soft = "y"`.
- Data types: `src/calibration_data_module.f90`.

Purpose

Soft calibration adjusts a small set of HRU and plant parameters until the simulated average annual water balance and crop yields match targets given for each land-use group. The targets are soft data: regional estimates such as the fraction of precipitation that leaves as surface runoff or evapotranspiration, the share of streamflow that is baseflow, or an average county yield. They are not time series.

Soft calibration runs inside the normal `swatplus` executable. After the base simulation ends, SWAT+ changes one parameter at a time and reruns the whole simulation period after each change. Each rerun is labelled on screen (`first esco adj`, `cn3_swf adj`, `epco adj`, and so on). A water balance calibration adds up to 17 full simulations in mode `a` and 20 in mode `b`, and plant calibration up to 13 more. The calibrated values are written to new files (see Output files); the model inputs are left unchanged.

In 62.0.1 two components work: the HRU water balance (`hyd` switch) and crop yield (`plnt` switch). The other switches are read, but they do not adjust anything. The status column of the switch table gives the detail for each.

Run sequence

1. At start-up SWAT+ reads the seven soft calibration files. A missing file, or a `null` name in `file.cio`, gives an empty table and that part is skipped. Without `codes.sft` soft calibration is off.
2. Any changes in `calibration.cal` are applied (see Hard calibration), so soft calibration starts from the changed values.
3. The base simulation runs as usual. At the end of each simulated year every HRU's annual water balance is added to its land-use group, and each harvest adds its yield to the plant targets.
4. If any switch in `codes.sft` is on, soft calibration runs next. It first sets the `print.prt` codes back to their defaults, which are mostly off, and turns on the average annual basin water balance. The warm-up period (`nyskip`) is kept. Most output files therefore contain the base run, and `basin_wb_aa.txt` gains a record for each calibration run.
5. Each adjustment step changes one parameter for every HRU in each land-use group that is outside its tolerance, restores the starting state of the model, and reruns the full period. The steps are listed under Adjustment steps.
6. The calibrated parameters are written out.

Input files

The `chg` line of `file.cio` names nine files. The first two are the hard calibration files described in Parameter change files. The last seven belong to soft calibration:

Position on `chg` line	Default name	Earlier name	Content
3	`codes.sft`	`codes.cal`	component switches
4	`wb_parms.sft`	`ls_parms.cal`	limits for the water balance parameters
5	`water_balance.sft`	`ls_regions.cal`	water balance targets by land-use group
6	`ch_sed_budget.sft`	`ch_orders.cal`	channel erosion targets

Position on chg line	Default name	Earlier name	Content
7	ch_sed_parms.sft	ch_parms.cal	limits for the channel parameters
8	plant_parms.sft	pl_parms.cal	plant parameter start values and limits
9	plant_gro.sft	pl_regions.cal	crop yield targets

A file is only read when its name is on the chg line. A project written by SWAT+ Editor 4.0.1 can contain the .sft files while their names on the chg line are null. In that case edits to the files have no effect.

All files are free format. Values are separated by blanks, and names must not contain blanks or slashes. The first line of every file is a title and is skipped.

codes.sft

- Line 1: title.
- Line 2: header.
- Line 3: eight one-letter switches in the order below.

#	Column	Values	Status in 62.0.1
1	hyd	n, a, b, y	Works. a and b select the mode; y behaves as b.
2	landscape	n, y	Not usable. It is meant for simplified (hru-lte) HRUs, but its target table is never set up.
3	plnt	n, y	Works, crop yield targets only.
4	sed	n, y	Not usable, see Upland sediment.
5	nut	n, y	Nothing is calibrated. Setting y starts the soft calibration runs and changes nothing.
6	ch_sed	n, y	Channel results are summed, but no parameter is adjusted.
7	ch_nut	n, y	Nothing is calibrated.
8	res	n, y	Nothing is calibrated.

Soft calibration runs when hyd is anything other than n or any other switch is y.

Mode a calibrates each water balance component against a separate ratio of precipitation. Mode b uses only the water yield ratio and the baseflow share (wyr and bfr in water_balance.sft) and derives the other targets from them. The steps for each mode are under Adjustment steps.

Example (SWAT+ Editor 4.0.1 layout with hyd set to b):

```
codes.sft: written by SWAT+ editor v4.0.1 on 2026-09-13 21:18 for SWAT+ rev. 62
hyd           landscape     plnt          sed           nut           ch_sed        ch_nut        res
b             n             n             n             n             n             n             n
```

water_balance.sft

- Line 1: title.
- Line 2: number of regions.
- Line 3: header.
- For each region: a line with the region name and the number of land-use rows. If that number is above zero, a header line follows, then one target row per land-use group.

Use a single region. The HRU list of a region is filled only when the region is named basin or the file has one region, and in both cases the region contains every HRU. A second region gets no HRUs and is not calibrated. HRU areas are taken from ls_unit.ele (the bsn_frac of element i is used for HRU i).

Target rows:

#	Column	Units	Description
1	`name`	none	Land-use group. Matched to the `cal_group` column of `landuse.lum`. The name `basin` matches every HRU.
2	`surq_rto`	fraction of precipitation	Surface runoff / precipitation. Mode `a` target for `cn3_swf`. In mode `b` the value is replaced when the file is read by `wyr` × (1 − `bfr`).
3	`latq_rto`	fraction of precipitation	Lateral flow / precipitation. Mode `a` target for `latq_co`. In mode `b` it is replaced by 0.1 × `wyr` × `bfr`.
4	`perc_rto`	fraction of precipitation	Percolation below the soil profile / precipitation. Mode `a` target for `perco`. In mode `b` it is added to `wyr` to form the `esco` target.
5	`et_rto`	fraction of precipitation	Actual ET / precipitation. Mode `a` target for `esco` and `pet_co`. Not used in mode `b`.
6	`tileq_rto`	fraction	Read; not used as a target.
7	`pet`	mm	Read; not used.
8	`sed`	t/ha	Upland sediment yield target for the `sed` switch.
9	`wyr`	fraction of precipitation	Water yield / precipitation. Mode `b` target for `pet_co` and `esco`.
10	`bfr`	fraction of water yield	Share of water yield that is baseflow. Mode `b` only. Converted when the file is read to baseflow / precipitation (`wyr` × `bfr`), the target for `perco`.
11	`solp`	kg/ha	Read; not used.

The simulated values are built from the HRU annual water balance (the columns of `hru_wb_yr`), weighted by HRU area:

- precipitation: `precip`
- surface runoff: `surq_gen`
- lateral flow: `latq`
- percolation: `perc`
- ET: `et`
- water yield: `surq_gen + latq + perc + qtile`
- baseflow: `latq + perc + qtile`

An HRU is left out of these sums once it has received more than 10 mm of irrigation in total. Its parameters are still changed with the rest of its group. The ratios in a row are not checked to add up to 1.

Example written by SWAT+ Editor 4.0.1 for revision 62. The row contains mode `a` ratios. `wyr` and `bfr` are zero, so in mode `b` every target would be zero and nothing would be calibrated.

```
water_balance.sft: written by SWAT+ editor v4.0.1 on 2026-09-13 21:18 for SWAT+ rev. 62
1
name                    num
basin                     1
    name                  surq_rto      latq_rto      perc_rto        et_rto     tileq_rto           pet
    basin                  0.20000       0.10000       0.10000       0.50000       0.00000       0.90000
```

wb_parms.sft

- Line 1: title.
- Line 2: number of rows.
- Line 3: header.
- Then one row per parameter: `name chg_typ neg pos lo up`.

SWAT+ uses the rows by position and does not look at `name` or `chg_typ`. Keep the ten rows in the order below; renaming or reordering them changes which limits each step uses.

Row	name	Parameter changed	Used by	How it is changed
1	cn2	curve number of simplified HRUs	landscape switch; pos is also used by the sed switch	added
2	esco	esco in hydrology.hyd	modes a and b	added
3	latq_co	latq_co in hydrology.hyd	modes a and b	added
4	petco	pet_co in hydrology.hyd	modes a and b	multiplied
5	slope	HRU slope	sed switch	subtracted
6	tconc	time of concentration	sed switch	divided
7	etco	ET coefficient of simplified HRUs	landscape switch	added
8	perco	perco in hydrology.hyd	modes a and b	added
9	revapc	revap coefficient of simplified HRUs	landscape switch	added
10	cn3_swf	cn3_swf in hydrology.hyd, with the curve numbers recomputed	modes a and b	added

For the HRU water balance rows (2, 3, 4, 8, 10):

- `neg` and `pos` limit the adjustment made for one land-use group. For `petco` they limit the multiplier (Editor default 0.54 to 1.85). A group that reaches a limit is flagged, and some later runs skip it.
- `lo` and `up` bound the HRU parameter value after every change.

Mode `b` narrows the `petco` range for its first PET step. It halves `neg` and `pos` and moves `lo` and `up` halfway toward 1, then restores the full range for the second PET step. See Limits in 62.0.1 for what this does with the Editor defaults.

`cn3_swf` and `perco` are not changed on tile-drained HRUs. At start-up those HRUs are set to `cn3_swf` = 0.95 and `perco` = 0.1.

Example written by SWAT+ Editor 4.0.1 for revision 62:

```
wb_parms.sft: written by SWAT+ editor v4.0.1 on 2026-09-13 21:18 for SWAT+ rev. 62
10
name                      chg_typ          neg          pos           lo           up
cn2                        abschg     -6.00000      6.00000     35.00000     95.00000
esco                       abschg     -1.00000      1.00000      0.00000      1.00000
latq_co                    abschg     -1.00000      1.00000      0.00000      1.00000
petco                      abschg      0.54000      1.85000      0.70000      1.20000
slope                      pctchg    -25.00000     25.00000      0.00010      0.90000
tconc                      pctchg    -30.00000     30.00000      5.00000    960.00000
etco                       abschg     -0.40000      0.40000      0.80000      1.20000
perco                      abschg     -0.70000      0.70000      0.00100      1.00000
revapc                     abschg     -0.40000      0.40000      0.00000      0.40000
cn3_swf                    abschg     -1.20000      1.20000      0.00000      1.00000
```

plant_gro.sft

- Line 1: title.

- Line 2: number of regions.
- Line 3: header.
- For each region:
  - a line with `name`, the number of plants (`lum_num`), `nspu`, and then `nspu` HRU numbers;
  - if `lum_num` is above zero, a header line and one target row per plant.

With `nspu` = 0 the region covers every HRU. Otherwise `nspu` numbers follow, and a negative number closes a range: with `nspu` = 3, the list `1 -20 35` means HRUs 1 to 20 and HRU 35. Each HRU belongs to one plant region. `plant_parms.sft` is read second, so its region assignment is the one that applies.

Target rows:

#	Column	Units	Description
1	`name`	none	Plant name. Matched to the plant named in the harvest operation (`harv` or `hvkl` in a management schedule, `harvest` or `harvest_kill` in a decision table).
2	`yield`	t/ha	Target yield. Compared with the total harvested yield divided by the total harvested area over the whole run.
3	`npp`	t/ha	Read; not used.
4	`lai_mx`	none	Read; not used.
5	`wstress`	none	Read; not used.
6	`astress`	none	Read; not used.
7	`tstress`	none	Read; not used.

plant_parms.sft

- Line 1: title.
- Line 2: number of regions. Region i here is paired with region i of `plant_gro.sft`.
- Line 3: header.
- For each region:
  - a line with `name`, `lum_num`, `parms` and `nspu`;
  - if `lum_num` is above zero, a header line and `lum_num` × `parms` parameter rows.

Set `nspu` to 0 here, which puts every HRU in the region. With `nspu` above zero the region line is read a second time without the `parms` field, so the HRU list is misread.

Set `parms` to 4. The rows form four blocks of `lum_num` rows each, in this order: `pest_stress`, `epco`, `lai_pot`, `harv_idx`. Within every block the plants must be in the same order as in the matching `plant_gro.sft` region. SWAT+ finds a row by its position in the block and always runs all four steps.

Parameter rows:

#	Column	Description
1	`var`	`pest_stress`, `epco`, `lai_pot` or `harv_idx`
2	`name`	plant name
3	`init_val`	start value, applied when the file is read to every plant of that name in the region's HRUs. This happens whether or not `plnt` is on.
4	`chg_typ`	read; not used
5	`neg`	lower limit of the adjustment. For `epco` it is the lowest `epco` value the calibration may set.
6	`pos`	upper limit of the adjustment. For `epco` it is the highest `epco` value the calibration may set.
7	`lo`	lower bound on the plant's value

#	Column	Description
8	`up`	upper bound on the plant's value

ch_sed_budget.sft and ch_sed_parms.sft

These two files are read, but no channel parameter is adjusted.

`ch_sed_budget.sft` layout:

- Line 1: title.
- Line 2: number of regions.
- Line 3: header.
- For each region: `name`, the number of target rows (`ord_num`), `nspu` and `nspu` channel numbers (0 means all channels). A header line always follows, then `ord_num` rows of `name chw chd hc fpd`.

The row fields are channel widening (mm/yr), downcutting or accretion (mm/yr), headcut advance (m/yr) and floodplain accretion (mm/yr).

`ch_sed_parms.sft` has the same layout as `wb_parms.sft`. Only `name` and `chg_typ` of its first row are used, and only when writing the channel summary.

The channel results are only summed when `ch_sed` is the only switch on, and only target rows named `basin` collect them.

Adjustment steps

In every step:

- The tolerance is the relative difference |target − simulated| / target, taken over the average annual values of the land-use group.
- A step only reruns the simulation if at least one group is outside tolerance.
- The first run of a step moves the parameter by a fixed rate. Later runs use a secant estimate from the last two runs.

Water balance, mode a

Step	Parameter	Target compared	Tolerance	Runs	First-run change	`wb_parms.sft` row
1	`esco`	`et_rto` × precipitation vs simulated ET	1 %	1 + 2	+ (target − simulated, mm) / 500	2
2	`pet_co`	same as step 1	1 %	2	multiplied by a ratio near 1	4
3	`cn3_swf`	`surq_rto` × precipitation vs `surq_gen`	2 %	1 + 2	− (target − simulated, mm) / 100	10
4	`latq_co`	`latq_rto` × precipitation vs `latq`	10 % first run, 5 % after	1 + 1	+ 0.0025 × (target − simulated, mm)	3
5	`perco`	`perc_rto` × precipitation vs `perc`	5 %	1 + 3	+ 0.001 × (target − simulated, mm)	8
6	`cn3_swf`	as step 3	2 %	1 + 2	as step 3	10

For example, with `surq_rto` = 0.2, 800 mm of precipitation and 120 mm of simulated surface runoff, the target is 160 mm, the difference is 40 mm, and the first `cn3_swf` run lowers `cn3_swf` by 0.4.

Water balance, mode b

Step	Parameter	Target compared	Tolerance	Runs
1	`pet_co`, half range	`wyr` × precipitation vs simulated water yield	1 %	2
2	`esco`	(`wyr` + `perc_rto`) × precipitation vs simulated water yield	1 %	1 + 2
3	`pet_co`, full range	as step 1	1 %	2
4	`cn3_swf`	`wyr` × (1 − `bfr`) × precipitation vs `surq_gen`	2 %	1 + 2
5	`latq_co`	0.1 × `wyr` × `bfr` × precipitation vs `latq`	10 % first run, 5 % after	1 + 2
6	`perco`	`wyr` × `bfr` × precipitation vs `latq` + `perc` + `qtile`	5 %	1 + 3
7	`cn3_swf`	as step 4	2 %	1 + 2

Crop yield

Each plant's average harvested yield is compared with its `plant_gro.sft` target:

1. Aeration stress. If the yield is more than 20 % below target, HRUs growing that plant that have no tile drain and an accumulated aeration stress above 50 get `perco` = 0.95. One run.
2. `epco`, with a 3 % tolerance. The first run sets `epco` to 1, or to a low value when the yield is more than 10 % above target. Four more runs narrow the interval between 0 and 1. The value is kept within `neg` and `pos` of the `epco` block.
3. `pest_stress`, changed by the relative yield error. One run.
4. `lai_pot`, 1 LAI unit for every 20 % of yield difference, then secant. 1 + 2 runs.
5. `harv_idx`, half the relative yield difference, then secant. 1 + 2 runs.

Upland sediment

The `sed` switch is meant to fit the `sed` targets by shortening the time of concentration and lowering the HRU slope, with the USLE slope-length factor recomputed. In 62.0.1 it does not work:

- Its loops run over the number of channel regions instead of the number of water balance regions.
- Simulated sediment is only summed when `hyd` is also on.
- The secant runs always compute a change of zero.

Treat the `sed` switch as unavailable.

Output files

File	Written when	Content
`hydrology-cal.hyd`	`hyd` or `plnt` on	One row per HRU with the calibrated `hydrology.hyd` values, written at the end.
`plant_parms.cal`	`plnt` on	Same layout as `plant_parms.sft`, with `init_val` updated to the calibrated values and `nspu` set to 0.
`hru-out.cal`	any switch on	Header only, except with the `landscape` switch, which appends the simplified HRU parameters.
`hru-new.cal`	`hyd` on	Opened but left empty.

To use the results in later runs, point file.cio at the new files:

- Use plant_parms.cal as plant_parms.sft. Its init_val values are applied when the file is read, also with plnt = n.
- Use hydrology-cal.hyd as hydrology.hyd, after the edits in the warning below. Rows are written per HRU under the name of its hydrology.hyd record, so every HRU needs a separate record. The extra 16th column (perco_lim, labelled NOT_USED) is ignored when the file is read.

  Converting hydrology-cal.hyd back to an input

  Two edits are needed before hydrology-cal.hyd can replace hydrology.hyd:

  - LAT_TTIME contains the value after the start-up conversion, 1 − exp(−1 / days). Read back as an input it is taken as days and converted a second time, which makes lateral flow drain much faster. Put the original lat_ttime values back before using the file.
  - When hyd and plnt are both on, the file contains the title, header and HRU rows twice. Keep only the second set, which is written after plant calibration.

Limits in 62.0.1

- Only one water balance region works (see water_balance.sft).
- wb_parms.sft and plant_parms.sft rows are found by position, so the row order is part of the format.
- In mode b the first PET step halves neg and pos for petco. With the Editor defaults (0.54, 1.85) the upper limit becomes 0.925. When simulated water yield is too high the step should raise PET; it caps the multiplier at 0.925 and lowers PET instead.
- plant_parms.sft must use nspu = 0 and parms = 4.
- The landscape, sed, nut, ch_sed, ch_nut and res switches do not calibrate anything (see codes.sft).
- Annual sums include the warm-up years, but the averages divide by the printed years only. Target ratios are unaffected because precipitation is treated the same way. The fixed first-run rates, which are in mm, act on values scaled by the same factor.

Related pages

- Hard calibration. Changes from calibration.cal, applied before the base run.
- Parameter change files. cal_parms.cal and calibration.cal, the first two files on the chg line.
- landuse.lum. The cal_group column that water_balance.sft rows are matched against.
- hydrology.hyd. The parameters written to hydrology-cal.hyd.

  Sources

  - Readers, called from src/proc_cal.f90 in this order: pl_read_regions_cal.f90 (plant_gro.sft), pl_read_parms_cal.f90 (plant_parms.sft), calsoft_read_codes.f90 (codes.sft), lcu_read_softcal.f90 (water_balance.sft), ls_read_parms_cal.f90 (subroutine ls_read_lsparms_cal, wb_parms.sft), ch_read_orders_cal.f90 (ch_sed_budget.sft), ch_read_parms_cal.f90 (ch_sed_parms.sft).
  - Control: src/calsoft_control.f90, called from src/main.f90.in after the base simulation when cal_soft = "y".
  - Water balance: src/calsoft_hyd.f90 (mode a); src/calsoft_hyd_bfr.f90 and its steps calsoft_hyd_bfr_pet, _et, _surq, _latq, _perc (mode b).
  - Crop yield: src/calsoft_plant.f90, calsoft_plant_zero.f90, pl_write_parms_cal.f90. Yields are added to the targets at harvest in mgt_sched.f90 and actions.f90.
  - Upland sediment: src/calsoft_sed.f90. Simplified HRU water balance: caltsoft_hyd.f90.
  - Annual sums and averages: calsoft_sum_output.f90 (end of every simulated year, called from time_control.f90) and calsoft_ave_output.f90 (end of every run).
  - Tile-drained HRU settings: topohyd_init.f90. lat_ttime conversion: hydro_init.f90. Output files are opened in header_write.f90.
  - Types: src/calibration_data_module.f90. File names: type input_chg in src/input_file_module.f90.

File and column	Fortran variable
codes.sft hyd, landscape, plnt, sed, nut, ch_sed, ch_nut, res	cal_codes%hyd_hru, %hyd_hrul, %plt, %sed, %nut, %chsed, %chnut, %res
water_balance.sft target row	type soft_calib_ls_processes: name, srr, lfr, pcr, etr, tfr, pet, sed, wyr, bfr, solp
wb_parms.sft rows	ls_prms(1:10)
plant_gro.sft target row	type soft_calib_pl_processes
plant_parms.sft parameter row	type pl_parms_cal; row k of block b is prm(k + (b-1)*lum_num)
HRU land-use group	hru%lum_group_c, from landuse.lum cal_group
accumulated aeration stress	strsa

Output file	Unit
hydrology-cal.hyd	5001
plant_parms.cal	107
hru-out.cal	4999
hru-new.cal	5000

Purpose

Hard calibration fits the model to measured time series such as streamflow, sediment or nutrient loads at a gauge. SWAT+ does not search for parameter values itself. An external tool (SWAT+ Toolbox, SWAT-CUP, a script) picks a set of parameter changes, writes them to calibration.cal, runs SWAT+, compares the output with the observations, and repeats. Each run of SWAT+ applies the changes in calibration.cal once, at start-up, and then simulates the full period with the changed parameters.

The input files are never edited. A change exists only for the run that reads it, so the same TxtInOut folder can be reused for every trial.

Files

Two files, both named on the chg line of file.cio, drive hard calibration:

- cal_parms.cal lists every parameter that may be changed, the object type it belongs to, and the lowest and highest value it may take. SWAT+ Editor writes this file and it rarely needs editing.
- calibration.cal is the change list for one run. Each row names a parameter from cal_parms.cal, the type and size of the change, and optionally the objects to change and the conditions they must meet.

If either slot is null or the file is missing, no changes are applied. The column layouts and the full list of parameter names are on Parameter change files.

Run sequence

1. SWAT+ reads all input files and sets up every object (HRUs, soils, plants, channels, aquifers, reservoirs).
2. It reads cal_parms.cal and calibration.cal.
3. For each row of calibration.cal, in file order, it goes through the listed objects (all objects of that type when none are listed), checks the row's conditions for each object, and changes the parameter on every object that meets them.
4. The simulation runs once with the changed parameters.

Because the changes are applied after the inputs are read and before the first day is simulated, they act for the whole simulation period. The year and day columns of calibration.cal only limit changes to the precipitation and temperature records (see Climate changes).

Rows are applied in order, and a later row for the same parameter acts on the value left by the earlier row. Two pctchg rows of 10 for cn2 give a 21 % increase.

Change types

Type	New value	Example (current value 70, change 5)
`absval`	the change value replaces the current value	5
`abschg`	current value + change	75
`pctchg`	current value × (1 + change / 100)	73.5
`relchg`	current value × (1 + change)	420

After every change the result is limited to the range `abs_min` to `abs_max` given for that parameter in `cal_parms.cal`. Use one of the four types; any other text in the change type column gives an undefined value.

Object types

The `obj_typ` column of `cal_parms.cal` tells SWAT+ which objects a parameter belongs to. When a row of `calibration.cal` lists no objects, every object of that type is changed.

Type	Objects changed when none are listed	Example parameters
`hru`	every HRU	`cn2`, `esco`, `epco`, `perco`, `lat_ttime`, `canmx`
`sol`	every HRU, in the soil layers from `lyr1` to `lyr2`	`awc`, `k`, `bd`, `usle_k`
`plt`	see Plant parameters	`lai_pot`, `harv_idx`, `phu_mat`
`hlt`	every HRU of the simplified `hru-lte` type	`cn2_lte`, `awc_lte`
`aqu`	every aquifer	`alpha`, `revap_co`, `flo_min`
`sdc`	every channel	`chn`, `chk`, `bank_exp`
`swq`	every channel nutrient parameter set	`rs1`, `bc1`, `mumax`
`res`	every reservoir	`k_res`, `evrsv`, `nsed`
`rdt`	every reservoir decision table	`drawdown_days`, `withdraw_rate`
`bsn`	the basin (applied once)	`surlag`, `cdn`, `nperco`
`gwf`	every gwflow cell	`aquifer_K`, `aquifer_Sy`
`gwf_riv`	every channel connected to gwflow	`stream_K`, `stream_thk`
`gwf_sgl`	applied once	`stream_bed`, `reservoir_K`
`cli`	see Climate changes	`precip`, `temp`

Soil layers

For `sol` parameters, `lyr1` and `lyr2` in `calibration.cal` set the first and last soil layer to change. A `lyr1` of 0 starts at the top layer, and a `lyr2` of 0 or more than the number of layers ends at the bottom layer.

Reservoir decision tables

For `rdt` parameters (`drawdown_days`, `withdraw_rate`) the five integer columns after `conds` select which storage zones of the decision table to change. A non-zero value in the column switches that zone on:

Column	Zone
`lyr1`	exclusive flood control storage
`lyr2`	seasonal flood control storage, non-flood season
`year1`	seasonal flood control storage, flood season
`year2`	multiple use storage, non-flood season
`day1`	multiple use storage, second zone

Plant parameters

Plant parameters with object type `plt` are changed in the plant community of each listed HRU, for the plants named in the row's conditions. Only `phu_mat`, `lai_pot` and `harv_idx` are handled this way. Each row needs a condition that names the plant (see Conditions and regionalization); a row without one changes nothing.

Give the HRUs to change in the object list of the row. When the list is empty, SWAT+ goes through as many HRUs as there are plants in plants.plt, starting from HRU 1.

To change epco use object type hru. That changes the HRU value and sets the same value for every plant in the HRU's community. With object type plt, an epco row changes lai_pot instead.

The plant carbon parameters bio_e, hvsti, rdmx, t_opt and t_base are listed with object type plt in the SWAT+ Editor cal_parms.cal. In 62.0.1 rows for these parameters change nothing.

Climate changes

Precipitation and temperature records can be changed over a date window with the parameters precip and temp, giving object type cli in cal_parms.cal. The change is applied to every daily value from day day1 to day day2 of each year from year1 to year2 (calendar years, inside the simulation period).

List the records to change in the object list of the row. With no list, nothing is changed. List one record per row. When a row lists several records, the change is applied to every listed record once for each record in the list, so a pctchg of 10 on three records gives a 33 % increase on each. In 62.0.1 temp rows do not change the listed temperature record.

Unknown names

A row whose name is not in cal_parms.cal, or whose name SWAT+ does not recognise as a parameter, is skipped without a message. Check spelling against the parameter list if a change seems to have no effect.

Related pages

- Parameter change files. Column layout of cal_parms.cal and calibration.cal, and the parameter names.
- Conditions and regionalization. Limiting a change to objects that meet a condition.
- Sensitivity workflow. Driving many runs from an external sampler.
- Soft calibration. Fitting the average annual water balance and crop yields before hard calibration.

Sources

- Readers: src/cal_parm_read.f90 (cal_parms.cal) and src/cal_parmchg_read.f90 (calibration.cal), called from src/proc_cal.f90.
- Applying changes: src/cal_conditions.f90, which calls src/cal_parm_select.f90 for each object. The change operator is src/chg_par.f90.
- Order of calls: src/main.f90.in. proc_cal runs after proc_hru, proc_cha, proc_aqu and the reservoir and wetland setup, and before time_control.
- Types: type calibration_parameters and type update_parameters in src/calibration_data_module.f90.

Purpose

cal_parms.cal is the list of parameters that calibration may change: the name of each parameter, the type of object it belongs to, and the lowest and highest value it may take. calibration.cal is the change list for one run: each row picks a parameter from cal_parms.cal, says how to change it, and optionally limits the change to some objects or to objects that meet conditions. How SWAT+ applies the list is described on Hard calibration.

Both files are named on the chg line of file.cio, in the first two positions (defaults cal_parms.cal and calibration.cal). A null name or a missing file switches the change list off.

All values are free format and separated by blanks. Names must not contain blanks.

cal_parms.cal

- Line 1: title (skipped).
- Line 2: number of parameter rows.

- Line 3: column header (skipped).
- Then one row per parameter.

#	Column	Description
1	`name`	Parameter name, as listed under Parameter names.
2	`obj_typ`	Type of object the parameter belongs to: `hru`, `sol`, `plt`, `hlt`, `aqu`, `sdc`, `rte`, `swq`, `res`, `rdt`, `bsn`, `pst`, `gwf`, `gwf_riv`, `gwf_sgl` or `cli`. It decides which objects are changed when a row of `calibration.cal` lists none.
3	`abs_min`	Lowest value the parameter may take. Every change is limited to this value.
4	`abs_max`	Highest value the parameter may take. Every change is limited to this value.
5	`units`	Units, for information only.

SWAT+ Editor writes this file with suitable limits for each parameter, and it rarely needs editing. A narrower range for a calibration run belongs in the calibration tool; the range here is the physical limit.

Example from `refdata/Osu_1hru` in the SWAT+ repository:

```
cal_parms.cal: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
186
name                      obj_typ       abs_min       abs_max            units
cn2                           hru      35.00000      95.00000             null
cn3_swf                       hru       0.00000       1.00000             null
usle_p                        hru       0.00000       1.00000             null
ovn                           hru       0.01000      30.00000             null
elev                          hru       0.00000    5000.00000                m
slope                         hru       0.00010       0.90000              m/m
slope_len                     hru      10.00000     150.00000                m
lat_ttime                     hru       0.50000     180.00000             days
```

The count on line 2 must match the number of rows. Rows past the count are not read.

calibration.cal

- Line 1: title (skipped).
- Line 2: number of change rows.
- Line 3: column header (skipped).
- Then one row per change, each followed directly by its condition lines, if it has any.

Change rows

#	Column	Description
1	`name`	Parameter name. Must also be in `cal_parms.cal`.
2	`chg_typ`	`absval` (replace), `abschg` (add), `pctchg` (percent change) or `relchg` (relative change). See Change types.
3	`val`	Size of the change.
4	`conds`	Number of condition lines that follow this row. 0 for none.
5	`lyr1`	First soil layer to change, for `sol` parameters (0 = top layer). For reservoir decision tables, a zone switch.
6	`lyr2`	Last soil layer to change, for `sol` parameters (0 = bottom layer). For reservoir decision tables, a zone switch.
7	`year1`	First year of the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
8	`year2`	Last year of the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
9	`day1`	First day of the year in the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.

#	Column	Description
10	`day2`	Last day of the year in the window, for `precip` and `temp`.
11	`obj_tot`	Number of entries in the object list that follows on the same line. 0 means every object of the parameter's type.
12…	object list	`obj_tot` object numbers.

The object numbers are the sequence numbers of the objects (the first HRU is 1, and so on). A negative number closes a range started by the number before it, so `obj_tot` = 3 with the list `1 -20 35` means objects 1 to 20 and object 35.

Condition lines

Each change row is followed by `conds` condition lines. A condition line has one of two forms:

- `range <var> <min> <max>`: the object's value of `<var>` must lie between `<min>` and `<max>`. Only `res_pvol` is recognised.
- `<var> <alt> <targ> <targc>`: a test against the text value `<targc>`. `<alt>` and `<targ>` must be present but are not used by the conditions that exist.

The variables and how they are tested are on Conditions and regionalization.

Example

`refdata/Osu_1hru/calibration.cal`:

```
  hru-new.cal developed from soft data calibration
    13
 NAME           CHG_TYP                  VAL   CONDS  LYR1  LYR2   YEAR1  YEAR2   DAY1   DAY2  OBJ_TOT
cn2         pctchg                 7.714000E+00  0          0          0          0          0          0
petco       pctchg                -5.273950E+01  0          0          0          0          0          0
latq_co     pctchg                -3.732030E+01  0          0          0          0          0          0
alpha       pctchg                 6.859000E+00  0          0          0          0          0          0
epco        pctchg                 6.534100E+01  0          0          0          0          0          0
awc         pctchg                -6.256550E+01  0          0          0          0          0          0
esco        pctchg                 3.305500E+01  0          0          0          0          0          0
perco       pctchg                 1.434000E+00  0          0          0          0          0          0
cn3_swf     pctchg                -7.710000E-02  0          0          0          0          0          0
canmx       pctchg                -6.680230E+01  0          0          0          0          0          0
lat_len     pctchg                -6.241200E+00  0          0          0          0          0          0
revap_co    pctchg                -5.854530E+01  0          0          0          0          0          0
surlag      pctchg                -2.125720E+01  0          0          0          0          0          0
```

Every row has no conditions and an empty object list, so each change applies to all objects of its type. The rows have one more zero than the header has columns; the reader stops after `obj_tot` when it is 0, so the extra value is ignored.

A row that changes `cn2` by +5 only on HRUs 1 to 20 and HRU 35, and only where the hydrologic soil group is `C`:

```
cn2         abschg      5.0    1   0   0   0   0   0   0   3   1  -20   35
hsg         =           0      C
```

Parameter names

The names SWAT+ 62.0.1 recognises are listed below with the object type to give in `cal_parms.cal` and the input the parameter comes from. A name that is in neither list is skipped without a message. Some recognised names have no effect at 62.0.1, or change a different value; the notes with each name say which.

HRU parameters (`hru`)

Name	Input	Notes
`cn2`	condition II curve number from `cntable.lum`	The curve numbers for the other moisture conditions are recomputed.
`cn3_swf`	`cn3_swf`, `hydrology.hyd`	Not changed on tile-drained HRUs.
`perco`	`perco`, `hydrology.hyd`	Not changed on tile-drained HRUs.
`esco`, `epco`	`esco`, `epco`, `hydrology.hyd`	`epco` is also set for every plant in the HRU.
`petco`	`pet_co`, `hydrology.hyd`	
`latq_co`, `lat_sed`	`latq_co`, `lat_sed`, `hydrology.hyd`	
`lat_ttime`	`lat_ttime`, `hydrology.hyd`	Use `absval` with a value in days. `abschg`, `pctchg` and `relchg` act on an internal coefficient and give wrong travel times.
`canmx`	`can_max`, `hydrology.hyd`	
`biomix`	`bio_mix`, `hydrology.hyd`	
`erorgn`, `erorgp`	`orgn_enrich`, `orgp_enrich`, `hydrology.hyd`	
`lat_orgn`, `lat_orgp`	`lat_orgn`, `lat_orgp`, `hydrology.hyd`	
`slope`, `slope_len`, `lat_len`, `dis_stream`	`slp`, `slp_len`, `lat_len`, `dist_cha`, `topography.hyd`	
`elev`	HRU elevation	
`ovn`	Manning's n for overland flow, `ovn_table.lum`	
`usle_p`	USLE support practice factor, `cons_practice.lum`	
`field_len`, `field_wid`, `field_ang`	`len`, `wd`, `ang`, `field.fld`	
`snofall_tmp`, `snomelt_tmp`, `snomelt_max`, `snomelt_min`, `snomelt_lag`	`fall_tmp`, `melt_tmp`, `melt_max`, `melt_min`, `tmp_lag`, `snow.sno`	
`tile_dep`, `tile_dtime`, `tile_lag`, `tile_rad`, `tile_dist`, `tile_drain`, `tile_pump`	`dp`, `t_fc`, `lag`, `rad`, `dist`, `drain`, `pump`, `tiledrain.str`	
`tile_latk`	`lat_ksat`, `tiledrain.str`	The result is written to the tile lag time (`lag`).
`cmn_h`, `nperco_h`, `pperco_h`, `phoskd_h`, `psp_h`, `nperco_lchtile_h`	HRU values of the basin nutrient parameters `cmn`, `nperco`, `pperco`, `phoskd`, `psp` and `nperco_lchtile`	Change one HRU's copy. The basin names below change the basin value.

Soil layer parameters (`sol`)

Changed in soil layers `lyr1` to `lyr2` of each HRU.

Name	Input	Notes
`z`	layer depth `dp`, `soils.sol`	
`bd`, `awc`, `k`	`bd`, `awc`, `soil_k`, `soils.sol`	Dependent water contents are recomputed after `awc` changes.
`clay`, `silt`, `sand`, `rock`	`clay`, `silt`, `sand`, `rock`, `soils.sol`	
`alb`, `usle_k`, `ec`, `cal`, `ph`	`alb`, `usle_k`, `ec`, `caco3`, `ph`, `soils.sol`	
`anion_excl`, `crk`	`anion_excl`, `perc_crk`, `soils.sol`	Whole profile values.
`cbn`	`carbon`, `soils.sol`	No effect: the soil carbon pools are set up after the change list is applied.
`lab_p`, `hum_c_n`, `hum_c_p`	`lab_p`, `hum_c_n`, `hum_c_p`, `nutrients.sol`	No effect, for the same reason.

Plant parameters (`plt`)

Changed for the plants named in the row's `plant` conditions, in the plant community of each listed HRU. See Plant parameters for how to write these rows.

Name	Input
`lai_pot`	`lai_pot`, `plants.plt`

Name	Input
harv_idx	harv_idx, plants.plt
phu_mat	heat units to maturity of the plant in the HRU's community, computed at start-up from the plant and the climate

bio_e, hvsti, rdmx, t_opt, t_base and usle_c are recognised names, but rows for them change nothing.

Basin parameters (bsn)

Changed once for the whole basin.

Name	Input
surlag, prf, evrch, percop, cdn, sdnco, tb_adj, n_updis, rsd_covco	parameters.bsn
init_seq, init_microb, init_slow, init_passive, koc_c, solc_ratio, manure_c_frac, bio_consol, till_consol, sfc_rsd_photodeg, n_act_frac, cnr_cap, cnr_ref, cpr_cap, cpr_ref, t_cbn_min, t_cbn_opt, t_cbn_max	carbon.bsn
bmix_a, bmix_b, bmix_c, tillmix_a, tillmix_b, tillmix_c	biological and tillage mixing curve coefficients, carbon.bsn

Basin rows for adj_pkr, ffcb, dorm_hr, cmn, nperco, pperco, phoskd, psp, nperco_lchtile, rsdco, msk_co1, msk_co2, msk_x, plaps, tlaps and evlai change nothing. SWAT+ copies these basin values to the HRUs, channels, plants and weather stations while it sets them up, before the change list is applied, and does not read the basin value again. For the nutrient parameters, use the HRU names (cmn_h, nperco_h, pperco_h, phoskd_h, psp_h, nperco_lchtile_h) instead.

The per-layer carbon rates of carbon_lyr.bsn (hp_rate, hs_rate, microb_rate, meta_rate, str_rate, microb_top_rate, hs_hp, a1co2, asco2, apco2, abco2) are recognised names, but rows for them change nothing.

Aquifer parameters (aqu)

Name	Input
flo_init_mm, dep_bot, dep_wt_init, no3_init, minp_init, cbn_init, flo_dist, bf_max, alpha, revap_co, deep_seep, sp_yld, hlife_n, flo_min, revap_min	gw_flo, dep_bot, dep_wt, no3_n, sol_p, carbon, flo_dist, bf_max, alpha_bf, revap, rchg_dp, spec_yld, hl_no3n, flo_min, revap_min, aquifer.aqu

Channel parameters (sdc or rte)

Name	Input
chw, chd, chs, chl, chn, chk, cov, d50, ch_clay, carbon, ch_bd, chss, bank_exp, vcr_coef, bankfull_flo, fps, fpn, hc_kh, hc_hgt, hc_ini, ch_n_conc, ch_p_conc, ch_p_bio, pk_rto, fp_inun_days, n_setl, p_setl, n_sol_part, p_sol_part, n_dep_enr, p_dep_enr, arc_len_fr, bed_exp, wash_bed_fr	channel geometry, hydraulics, sediment and nutrient parameters of the SWAT-DEG channel inputs

Channel nutrient parameters (swq)

Name	Input
mumax, rs1 to rs7, rk1 to rk6, bc1 to bc4	in-stream nutrient and algae rates, nutrients.cha (see channel inputs)

A rk5 row changes rs2 (the ben_disp column of nutrients.cha) instead.

Reservoir parameters (res)

Name	Input
esa, evol, psa, pvol	emergency and principal spillway surface areas and volumes, hydrology.res
k_res, evrsv	bottom hydraulic conductivity and evaporation coefficient, hydrology.res
nsed, res_d50, sed_stlr, velsetlr	normal sediment concentration, median particle size, settling rate and settling velocity, sediment.res
psetlr1, psetlr2, nsetlr1, nsetlr2, nsolr, psolr, soln_stl_fr, solp_stl_fr	nutrient settling rates, nutrients.res
vol, sed, orgn, orgp, no3, nh3, no2, solp	water, sediment and nutrient content at the start of the run, initial.res

Reservoir decision table parameters (rdt)

Name	Input
drawdown_days, withdraw_rate	release rules of the reservoir decision tables in res_rel.dtl, per storage zone (see Reservoir decision tables)

Pesticide parameters (pst)

List the pesticides to change (their order in pesticide.pes) in the object list. With an empty list nothing is changed.

Name	Input
pst_koc, pst_washoff, pst_foliar_hlife, pst_soil_hlife, pst_solub, pst_aq_hlife, pst_aq_volat, pst_aq_settle, pst_aq_resus, pst_ben_hlife, pst_ben_bury, pst_ben_act_dep	soil_ads, frac_wash, hl_foliage, hl_soil, solub, aq_hlife, aq_volat, aq_settle, aq_resus, ben_hlife, ben_bury, ben_act_dep, pesticide.pes

Simplified HRU parameters (hlt)

Name	Input
cn2_lte, awc_lte, etco_lte, tc_lte, soildep_lte, slope_lte, slopelen_lte, sy_lte, abf_lte, revapc_lte, percc_lte, sw_lte, gw_lte, gwflow_lte, gwdeep_lte, snow_lte, uslek_lte, uslec_lte, uslep_lte, uslels_lte	hru-lte.hru

awc_lte is given per mm of soil depth; SWAT+ multiplies the change and the upper limit by the soil depth.

gwflow parameters

These only act when gwflow is switched on in codes.bsn. Each name needs the object type shown.

Name	Object type	Objects changed	Input
aquifer_K, aquifer_Sy, aquifer_exdp	gwf	gwflow cells	aquifer hydraulic conductivity, specific yield and extinction depth, gwflow inputs

Name	Object type	Objects changed	Input
`tile_K`, `floodplain_K`	`gwf`	gwflow cells	tile drain and floodplain conductivity, only when tile or floodplain exchange is on
`aquifer_delay`	`hru`	HRUs	recharge delay from the soil to the water table, in days
`stream_K`, `stream_thk`	`gwf_riv`	channels	streambed conductivity and thickness of the cells connected to each channel
`stream_bed`	`gwf_sgl`	applied once	streambed elevation offset, one value for all channels
`reservoir_K`, `pond_bed_K`	`gwf_sgl`	applied once	reservoir and pond bed conductivity

Climate parameters (`cli`)

Name	Input
`precip`	daily precipitation of the listed `*.pcp` records, from `day1` to `day2` of each year from `year1` to `year2`
`temp`	daily temperature of the listed `*.tmp` records; changes nothing

See Climate changes before using these.

Related pages

- Hard calibration. When and how the change list is applied.
- Conditions and regionalization. Condition line variables.
- Soft calibration. The seven `.sft` files that follow these two on the `chg` line.

Sources

- Readers: `src/cal_parm_read.f90` and `src/cal_parmchg_read.f90`. Object list expansion: `src/define_unit_elements.f90`.
- Parameter names and the variables they change: `src/cal_parm_select.f90`. Object type dispatch: `src/cal_conditions.f90`.
- Types: `type calibration_parameters`, `type calibration_conditions` and `type update_parameters` in `src/calibration_data_module.f90`. File name slots: `in_chg%cal_parms` and `in_chg%cal_upd` in `src/input_file_module.f90`.

Column	Fortran variable
`cal_parms.cal` `name`, `obj_typ`, `abs_min`, `abs_max`, `units`	`cal_parms(i)%name`, `%ob_typ`, `%absmin`, `%absmax`, `%units`
`calibration.cal` `name` to `day2`	`cal_upd(i)%name`, `%chg_typ`, `%val`, `%conds`, `%lyr1`, `%lyr2`, `%year1`, `%year2`, `%day1`, `%day2`
`calibration.cal` `obj_tot` and list	`cal_upd(i)%num_tot`, expanded into `cal_upd(i)%num(:)`
condition `var alt targ targc`	`cal_upd(i)%cond(j)%var`, `%alt`, `%targ`, `%targc`
condition `range var min max`	`cal_upd(i)%cond(j)%var`, `cal_upd(i)%val1`, `cal_upd(i)%val2` (shared by all range lines of the row)

Purpose

A row of calibration.cal can be followed by condition lines. SWAT+ then changes the parameter only on the objects that meet every condition of the row. Conditions are checked once, at start-up, against the state of each object after the inputs are read.

Conditions and object lists are the two ways to calibrate parts of a watershed separately, for example one set of curve number changes for cropland and another for forest, or different groundwater parameters upstream and downstream of a gauge.

Condition lines

The number of condition lines is given in the **conds** column of the change row, and the lines follow the row directly. Each line has one of two forms.

Range condition

`range <variable> <min> <max>`

The object's value must be greater than **<min>** and less than **<max>**.

Text condition

`<variable> <operator> <number> <text>`

The object's value must equal **<text>**. All four values must be present. The operator and number are read but none of the existing conditions uses them, so any placeholder works (for example **=** and **0**).

When a row has several condition lines, an object must meet all of them.

Condition variables

Variable	Form	Objects	Condition met when
cal_group	text	HRU	the **cal_group** of the HRU's land use in landuse.lum equals the text
landuse	text	HRU	the HRU's land use name (the **lu_mgt** entry of hru-data.hru) equals the text
hsg	text	HRU	the hydrologic soil group of the HRU's soil (**hyd_grp** in soils.sol) equals the text, for example **C**
texture	text	HRU	the **texture** of the HRU's soil in soils.sol equals the text
plant	text	HRU	the first plant of the HRU's plant community is the named plant
pl_class	text		see below
res_typ	text	reservoir decision table	the decision table's name equals the text
res_pvol	range	reservoir	the principal spillway volume, in 10^4 m^3 (ha-m, as in hydrology.res), is between **<min>** and **<max>**

The HRU conditions apply to parameters whose object type is **hru**, **sol** or **plt**. They compare against the HRU with the same number as the object being changed, so with other object types (aquifers, channels) they test an unrelated HRU.

Text comparisons are exact and case sensitive, and an unknown variable name is ignored, so a misspelt condition lets every object through.

Notes on specific variables:

- **plant**: only the first plant in the community is checked. For a community of corn and soybean in that order, **plant = corn** matches and **plant = soyb** does not. For the plant parameters **lai_pot**, **harv_idx** and **phu_mat** the **plant** condition has a second role: it names the plant whose value is changed (see Plant parameters).

- `pl_class`: compares the class of the plant at the same position in `plants.plt` as the object number. It does not look at the HRU's plants, so it is not useful for selecting HRUs.
- `res_pvol`: both limits are taken from the last `range` line of the row.

Regional calibration

Calibration groups

The most direct way to calibrate regions or land-use groups separately is the `cal_group` column of `landuse.lum`:

1. Give each land use a group name in `cal_group`, for example `crop_north`, `crop_south`, `forest`. Land uses that share a name form one group.
2. Write one change row per group, each with a `cal_group` condition.

```
cn2      pctchg   -8.0   1   0  0  0  0  0  0   0
cal_group  =  0  crop_north
cn2      pctchg    4.0   1   0  0  0  0  0  0   0
cal_group  =  0  forest
```

The same `cal_group` names are used by soft calibration to match its water balance targets to HRUs.

Object lists

The other way is to list the objects in the row itself (`obj_tot` and the numbers after it), for example the HRUs, aquifers or channels upstream of a gauge. A negative number closes a range, so `3  1 -120 245` selects objects 1 to 120 and 245. Object lists work for every object type.

Object lists and conditions can be combined: the conditions are then checked only on the listed objects.

Scheduled land use updates

SWAT+ also reads a file named `scen_dtl.upd` from the project folder, if it exists. It lists decision tables from the scenario decision table file (`scen_lu.dtl` by default, named on the `decision_table` line of `file.cio`) that are checked for every HRU at the start of every simulated day. Their actions can change land use, management or other HRU settings during the run, for example to simulate land use change at a set date.

- Line 1: title.
- Line 2: number of tables.
- Line 3: header.
- One line per table: maximum number of times the table may act, table type, and table name.

The maximum and the type are read but not used in 62.0.1: every listed table is checked every day for every HRU.

This file is separate from `calibration.cal`. Its changes happen during the run, while `calibration.cal` changes happen before the run starts. See Decision tables for how tables are written.

Related pages

- Parameter change files. Layout of the change rows.
- Hard calibration. When and how changes are applied.
- landuse.lum. The `cal_group` column.

Sources

- Reading condition lines: `src/cal_parmchg_read.f90`.
- Evaluating conditions: `src/cal_conditions.f90`.
- Calibration group of an HRU: `src/hrudb_init.f90` copies `lum(ilu)%cal_group` to `hru(ihru)%cal_group`.
- `scen_dtl.upd`: `src/cal_cond_read.f90`; used in `src/time_control.f90`.
- Types: `type calibration_conditions`, `type update_parameters` and `type update_conditional` in `src/calibration_data_module.f90`.

Condition variable	Compared with
cal_group	hru(ielem)%cal_group
landuse	hru(ielem)%land_use_mgt_c
hsg	soil(ielem)%hydgrp
texture	soil(ielem)%texture
plant	pcom(ielem)%pl(1)
pl_class	pl_class(ielem)
res_typ	dtbl_res(ielem)%name
res_pvol	res_ob(ielem)%pvol, against 10000 * val1 and 10000 * val2

Purpose

A sensitivity analysis shows which parameters change a chosen model result the most, so that calibration can concentrate on them. SWAT+ does not include a sensitivity analysis. It is run by an external tool that writes calibration.cal, runs SWAT+, reads a result from the output files, and repeats for many parameter sets. Each run applies the changes in calibration.cal at start-up and leaves the input files untouched, so one project folder serves every run.

SWAT+ Toolbox does this from a graphical interface. The steps below are what any tool or script has to do.

Steps

1. Choose the parameters to test and a range for each. The names are those on Parameter names. The abs_min and abs_max values in cal_parms.cal are physical limits; a narrower range around reasonable values usually gives more useful results.
2. Generate the samples: use a sampling design that matches the analysis method: Sobol (Saltelli sampling), FAST, Morris (elementary effects), Latin hypercube with one-at-a-time changes (LH-OAT), and so on. Each sample is one set of parameter values. Libraries such as SALib (Python) or sensitivity (R) provide both the sampling and the analysis.
3. For each sample, write calibration.cal with one row per parameter, run SWAT+ in the project folder, and compute the result of interest: total or peak flow at an outlet, a performance measure against observations (NSE, KGE, percent bias), a sediment or nutrient load.
4. Compute the indices: pass the sample matrix and the vector of results to the analysis method: first-order and total-order indices for Sobol and FAST, mean and spread of elementary effects for Morris.

The runs are independent, so they can be spread over several processor cores, each in a separate copy of the project folder. Sample sizes of several hundred to a few thousand runs are common.

Writing calibration.cal

A sample with three parameters, written as relative and absolute changes:

```
sensitivity sample 17
3
NAME      CHG_TYP      VAL    CONDS  LYR1  LYR2  YEAR1  YEAR2  DAY1  DAY2  OBJ_TOT
cn2       pctchg     -6.2     0      0     0     0      0      0     0     0
esco      absval      0.71    0      0     0     0      0      0     0     0
alpha     absval      0.048   0      0     0     0      0      0     0     0
```

Points to watch:

- pctchg scales each object's current value, so HRUs keep their differences. absval sets every object to the same value.
- Every parameter name must also be in cal_parms.cal, and its limits there still apply. A sample value outside abs_min to abs_max is cut to the limit, which flattens the response near the edges.

- A misspelt name is skipped without a message. Check a first run by comparing two extreme samples before starting the full set.
- Some parameter names have no effect in 62.0.1, or work only with restrictions; see the notes on Parameter change files and Hard calibration before including them.

Reading results

Daily channel output is in `channel_sd_day.txt` (or `.csv`), with flow out of each channel in the `flo_out` column (m^3/s). Switch on only the output the analysis needs in `print.prt`; writing every file for thousands of runs costs a lot of time and disk space. The output files are described in the Output Reference.

Related pages

- Parameter change files. Format of `cal_parms.cal` and `calibration.cal`, and the parameter names.
- Hard calibration. How each run applies the change list.

Sources

- The change list is read by `src/cal_parmchg_read.f90` and applied by `src/cal_conditions.f90`, both from `src/proc_cal.f90` before the simulation starts.

These pages use the two small datasets in the `refdata` folder of the SWAT+ source repository and go through them file by file. They assume SWAT+ is installed and the Getting started pages have been read.

Pages

- Ames_sub1 walkthrough. Twelve HRUs near Ames, Iowa, simulated from 1975 to 2020, with no routing between objects. Each input group in turn, the run, and the outputs.
- Osu_1hru walkthrough. One rice paddy HRU in South Korea, simulated from 2010 to 2023, with a routing unit, an aquifer and a channel. Paddy management by decision tables, and the shipped files that 62.0.1 reads differently from their headers.
- Carbon notebook. The Jupyter notebook `jupyter_notebooks/carbon.ipynb`, which plots the soil carbon output of one HRU, and how to run it with Ames_sub1.
- Cookbook recipes. Short edits: change the simulation period, turn on an output, change the PET method, switch to Green-Ampt, add an HRU, switch on the carbon model.

Sources

- Datasets: `refdata/Ames_sub1` and `refdata/Osu_1hru` at 62.0.1.
- Notebook: `jupyter_notebooks/carbon.ipynb` at 62.0.1.

`Ames_sub1` is the reference dataset used to test SWAT+. It has twelve HRUs near Ames, Iowa, and 46 years of daily simulation (1975 to 2020), with no channels, aquifers or reservoirs. The folder is `refdata/Ames_sub1` in the source repository.

The sections below go through the input files in the order of `file.cio`, then run the model and read the outputs, as a second pass after the Quick start. Several inputs of this test dataset do not do what their names suggest. They are pointed out where they come up, because they are also the mistakes to look for in a new project.

file.cio

`file.cio` lists every other input file. Most lines in Ames_sub1 are `null`:

```
simulation      time.sim          print.prt         null              object.cnt        null
basin           codes.bsn         parameters.bsn    carbon.bsn
climate         weather-sta.cli   weather-wgn.cli   null              pcp.cli
 ↪  tmp.cli
connect         hru.con
```

```
hru               hru-data.hru
hydrology         hydrology.hyd     topography.hyd
structural        tiledrain.str     septic.str        filterstrip.str   grassedww.str
 ↪  bmpuser.str
hru_parm_db       plants.plt        fertilizer.frt    tillage.til       pesticide.pes     ...
 ↪  snow.sno
ops               harv.ops          graze.ops         irr.ops           chem_app.ops
 ↪  fire.ops   sweep.ops
lum               landuse.lum       management.sch    cntable.lum       cons_practice.lum
 ↪  ovn_table.lum
init              plant.ini         soil_plant.ini
soils             soils.sol         nutrients.sol
decision_table    lum.dtl
regions           ls_unit.ele       ...  aqu_catunit.ele
```

The only spatial objects are HRUs (`hru.con`). There is no routing: each HRU's water, sediment and nutrients leave the model, and the results are read per HRU.

Reference: file.cio.

Simulation settings

time.sim

```
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

Day 0 at the start and end means 1 January of the first year and 31 December of the last. `step` 0 is a daily time step.

print.prt

```
nyskip      day_start  yrc_start  day_end   yrc_end   interval
0           0               1975         0     2020          1
...
objects                  daily       monthly        yearly         avann
hru_wb                       y             n             n             y
hru_nb                       y             n             n             y
hru_ls                       n             n             n             y
hru_pw                       y             n             n             y
hru_cb                       l             y             y             y
hru_cb_vars                  l             y             y             y
```

`nyskip` 0 means no warm-up years are left out of the outputs; for a real project set it to one to three years. Only HRU outputs are listed. Daily water balance, nutrient and plant output is on, and the carbon rows write daily output for each soil layer (`l`), which is what makes this run's output about 9 GB.

object.cnt

Twelve HRUs and no other objects.

References: time.sim, print.prt, object.cnt.

Basin settings

codes.bsn

```
pet_file  wq_file  pet  event  crack  swift_out  sed_det  rte_cha  deg_cha  wq_cha  nostress
 ↪  cn  c_fact  carbon  lapse  uhyd  sed_cha  tiledrain  wtable  soil_p  gampt  atmo_dep
 ↪  stor_max  i_fpwet  gw_flow  idc_till
null      null     2    0      0      0          0        0        0        0       0         2
 ↪  0       2       0      1     0        0          0       0       0      0         0
 ↪  0        0        3
```

- `pet` 2: potential evapotranspiration by the Hargreaves method.
- `cn` 2: read but not used in 62.0.1; the curve number always follows soil water.
- `carbon` 2: the soil carbon model is on, with its parameters from `carbon.bsn` and `carbon_lyr.bsn`.
- `gampt` 0: curve number runoff (Green and Ampt infiltration is off).

parameters.bsn and carbon.bsn

Basin-wide parameters and the carbon model coefficients. Ames_sub1 has `1.0` in column 7 of `parameters.bsn`, which switches on temperature-dependent pesticide decay.

References: codes.bsn, parameters.bsn, carbon.bsn.

Climate

`weather-sta.cli` defines one station:

```
name        wgn       pcp        tmp        slr   hmd   wnd   wnd_dir   atmo_dep
wgn_sta1    wgn_001   ames.pcp   ames.tem   sim   sim   sim   null      null
```

Precipitation comes from `ames.pcp`. Solar radiation, humidity and wind are generated (`sim`) from the monthly statistics in `weather-wgn.cli`.

Temperature does not come from a file. The station names `ames.tem`, but the temperature file list `tmp.cli` contains only `ames.tmp`, an empty file. The model cannot match the name, writes `ames.tem file not found (tgage)` to `diagnostics.out`, and generates daily temperatures from `weather-wgn.cli`. The measured record in `ames.tem` is never read. With Hargreaves PET, which depends on temperature, this affects ET as well.

`hru.con` gives each HRU the station name `wgn_01`, which is not the name in `weather-sta.cli` (`wgn_sta1`). With a single station the model uses it anyway, without a message; in a project with several stations, check that these names match.

References: weather-sta.cli, *.pcp, *.tmp, *.tem.

HRUs

hru-data.hru

Each row names the records an HRU uses:

```
id  name      topo          hydro     soil         lu_mgt            soil_plant_init  surf_stor
 ↪  snow     field
 1  hru0001   topohru0001   hyd0001   soil_01-h1   cosy_lum          soilplant1       null
 ↪  snow001  null
 2  hru0002   topohru0002   hyd0002   soil_02      mntill_corn_lum   soilplant1       null
 ↪  snow001  null
...
12  hru0012   topohru0012   hyd0012   soil_12      cosy_lum          soilplant1       null
 ↪  snow001  null
```

Eleven HRUs use `cosy_lum`, a corn and soybean rotation, and HRU 2 uses `mntill_corn_lum`, continuous corn with reduced tillage. Each HRU has a different soil.

hydrology.hyd and topography.hyd

HRU hydrology parameters (lateral flow travel time, `esco`, `epco`, `cn3_swf`, `perco`, `latq_co`, …) and slope, slope length and lateral flow length.

References: hru-data.hru, hydrology.hyd and topography.hyd.

Land use and management

`landuse.lum` links each land use to a management schedule:

Land use	Schedule in `landuse.lum`	Used by
`cosy_lum`	`mgt_01`	HRUs 1 and 3 to 12
`mntill_corn_lum`	`mgt_08`	HRU 2
`cosy_lum2`	`mgt_01A`	no HRU
`agrr_lum`, `soyb_lum`, `oats_lum`, `wwht_lum`	`AGRR_rot`, `SOYB_rot`, `OATS_rot`, `WWHT_rot`	no HRU; the schedules are not in `management.sch`

`management.sch` contains schedules `mgt_01` to `mgt_12` and `mgt_01A`, each a list of dated operations (tillage, planting, fertilizer, harvest). Two points about the shipped schedules at 62.0.1:

- The fertilizer name in the `fert` rows is one column to the right of where the model reads it, so the model finds no fertilizer and the rows apply nothing, without a message. The management page shows the expected layout.
- `lum.dtl` contains a decision table, `irr_year_irr`, meant to limit irrigation to 500 mm a year. At 62.0.1 the yearly total is not reset, so once 500 mm have been applied in total the table stops irrigation for the rest of the run.

References: landuse.lum, management.sch, *.dtl.

Databases and initial conditions

- `plants.plt`, `fertilizer.frt`, `tillage.til`, `pesticide.pes`, `urban.urb`, `septic.sep`, `snow.sno`: parameter databases.
- `harv.ops`, `graze.ops`, `irr.ops`, `chem_app.ops`, `fire.ops`, `sweep.ops`: parameters of each type of operation.
- `plant.ini`: the plant community each HRU starts with.
- `soil_plant.ini`: which nutrient record each HRU's soil starts with.
- `soils.sol` and `nutrients.sol`: soil layers and initial nutrient contents.

References: plants.plt, fertilizer.frt, tillage.til, *.ops, soils.sol, initial conditions.

Running the model

Check that there is about 10 GB of free space, or switch off the daily carbon output first (set the `daily` column of `hru_cb` and `hru_cb_vars` to `n` in `print.prt`). Then, from inside the folder:

```
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

The screen shows the date being simulated:

```
  Original Simulation            1   14   1994 Yr    20 of    46 Time   12:37:36
```

The run took 4 minutes 20 seconds on a recent Linux workstation, most of it spent writing the daily carbon files.

Outputs

simulation.out

The last lines should be:

```
  Original Simulation           12  31  2020 Yr   46 of   46 Time  12:40:12

 Execution successfully completed
```

If the completion line is missing, the run stopped early; the last date shows where.

diagnostics.out

```
 DIAGNOSTICS.OUT FILE
 agrr_lum      AGRR_rot      not found in management.sch
 soyb_lum      SOYB_rot      not found in management.sch
 oats_lum      OATS_rot      not found in management.sch
 wwht_lum      WWHT_rot      not found in management.sch
 ames.tem                    file not found (tgage)
```

The first four lines are land uses whose schedules are missing; no HRU uses them, so they do not affect the results. The last line is the temperature file described under Climate.

Reference: Checker and diagnostics.

checker.out

One row per HRU with the main soil and hydrology values the model ended up with:

```
 sname        hydgrp       zmx    usle_k    sumfc     sumul   usle_p  usle_ls   esco   epco
 ↪  cn3_swf  perco  latq_co  tiledrain
soil_01-h1    B         2000.0    0.0000  361.500   616.090   1.0000   0.3638 0.9500 1.0000
 ↪  0.0000 0.5000   0.0100          0
soil_02       B         2000.0    0.0000  361.500   449.038   0.7500   0.3638 0.9500 1.0000
 ↪  0.0000 0.5000   0.0100          0
soil_03-h3    B         2000.0    0.0000  361.500   610.323   1.0000   0.3638 0.9500 1.0000
 ↪  0.0000 0.5000   0.0100          0
...
```

`usle_k` is 0 for every soil, so the HRUs produce no sediment. Checking this file for zeros and implausible values is a quick test of any new project.

hru_wb_aa.txt

The average annual water balance of each HRU. For HRU 1:

precip	snofall	surq_gen	latq	perc	et
952.6 mm	110.3 mm	34.7 mm	0.1 mm	0.0 mm	932.5 mm

Almost all precipitation leaves as ET and nothing percolates. For a real Iowa watershed, ET would be closer to two thirds of precipitation. Comparing the average annual water balance with regional estimates is the first check of any project, and soft calibration adjusts it.

Columns: Water balance outputs.

Other outputs

- `hru_nb_*`, `hru_ncycle_*`: nutrient balance and nitrogen cycle per HRU.
- `hru_ls_aa.txt`: sediment and nutrient losses per HRU.
- `hru_pw_*`: plant growth and stress per HRU.
- `crop_yld_yr.txt`, `crop_yld_aa.txt`: yields by HRU and crop.
- `mgt_out.txt`: every management operation performed, by date and HRU. Use it to check that the schedules do what was intended.
- `hru_carbvars.txt`, `hru_cflux_stat.txt`, `hru_cpool_stat.txt` and the other carbon files: see Carbon outputs.

The shipped folder also contains `basin_*` files and others that this run does not write. They are left from earlier model versions; see Quick start.

Changes to try

- Set `nyskip` to 2 in `print.prt` and compare `hru_wb_aa.txt`.
- Rename `ames.tmp` to `ames.tem` in `tmp.cli` so that the measured temperatures are used, and compare ET.
- Change the PET method in `codes.bsn` (`pet` 0 for Priestley-Taylor, 1 for Penman-Monteith); both need solar radiation, humidity and wind, which are generated here. See the cookbook.
- Follow the Carbon notebook to plot the carbon output.

Sources

- Dataset: `refdata/Ames_sub1` at 62.0.1.
- Run: the 62.0.1 `gnu-lin_x86_64` release executable, 20 October 2026. `diagnostics.out`, `checker.out`, `hru_wb_aa.txt` and the run time are from that run.
- Fertilizer column: the `fert` case of `src/mgt_sched.f90` uses `mgt%op1`, the fertilizer matched from `op_data1` (column 5); the Ames `fert` rows have `null` there and the name in `op_data2`. Details on management.sch.
- Station and file name lookups, and the irrigation decision table: see weather-sta.cli and decision tables.

`Osu_1hru` is the second reference dataset in the SWAT+ source repository. It has one HRU of 10 ha, a rice paddy in South Korea with weather from the Imsil station, simulated from 2010 to 2023. Unlike Ames_sub1, the HRU belongs to a routing unit that sends its water to an aquifer and a channel, so the routing, aquifer and channel outputs have values. The folder is `refdata/Osu_1hru` in the source repository.

The rice is managed almost entirely by decision tables: weir height, paddy irrigation, puddling, planting, fertilizer and harvest are actions in `lum.dtl`. This makes the dataset a small example of paddy management. Several shipped files are in older formats or are not read by 62.0.1, and the sections below say what the model does with each one.

file.cio

```
simulation        time.sim          print.prt         null              object.cnt        null
basin             codes.bsn         parameters.bsn    null
climate           weather-sta.cli   weather-wgn.cli   null              pcp.cli
 ↪  tmp.cli           slr.cli           null              wnd.cli           null
connect           hru.con           null              rout_unit.con     null
 ↪  aquifer.con       null              null              null        recall.con
 ↪  null              null              null              chandeg.con
channel           initial.cha       null              null              null
 ↪  nutrients.cha     channel-lte.cha   hyd-sed-lte.cha   null
```

```
reservoir          initial.res       null              null              sediment.res
 ↪  nutrients.res      weir.res              wetland.wet       hydrology.wet
routing_unit       rout_unit.def     rout_unit.ele     rout_unit.rtu     null
hru                hru-data.hru      null
...
recall             recall.rec
...
aquifer            initial.aqu       aquifer.aqu
...
water_rights       null              null              null
...
hydrology          hydrology.hyd     topography.hyd    field.fld
structural         tiledrain.str     septic.str        filterstrip.str   grassedww.str
 ↪  bmpuser.str
hru_parm_db        plants.plt        fertilizer.frt    tillage.til       pesticide.pes     null
 ↪  null               null              urban.urb         septic.sep        snow.sno
ops                harv.ops          graze.ops         irr.ops           chem_app.ops
 ↪  fire.ops           sweep.ops
lum                landuse.lum       management.sch    cntable.lum       cons_practice.lum
 ↪  ovn_table.lum
chg                cal_parms.cal     calibration.cal   null ...
init               plant.ini         soil_plant.ini    om_water.ini      null ...
soils              soils.sol         nutrients.sol     null
decision_table     lum.dtl           res_rel.dtl       null              null
regions            ls_unit.ele       ls_unit.def       null ...          aqu_catunit.ele   null
 ↪  ...
```

The `basin` line already has three slots, with `null` for the carbon file. The `water_rights` line has three `null` names where 62.0.1 reads two, and the extra `null` is ignored. Both lines are read correctly (see older file.cio files).

The `chg` line names `calibration.cal`. Its 13 rows change `cn2`, `petco`, `latq_co`, `alpha`, `epco`, `awc`, `esco`, `perco`, `cn3_swf`, `canmx`, `lat_len`, `revap_co` and `surlag` by a percentage for every object, and the model applies these changes at start-up (see hard calibration). The run below includes them.

Reference: file.cio.

Simulation settings

time.sim

```
day_start  yrc_start   day_end   yrc_end      step
        1       2010       365      2023         0
```

1 January 2010 to 31 December 2023, 14 years, with a daily time step.

print.prt

```
nyskip      day_start  yrc_start  day_end   yrc_end   interval
3           0          0          0         0         1
aa_int_cnt
0
csvout         dbout          cdfout
n              n              n
soilout        mgtout         hydcon         fdcout
n              y              n              n
objects                  daily       monthly        yearly         avann
basin_wb                    n             n             y             y
basin_nb                    n             n             n             y
```

```
...
hru_pw                          n             n             n             y
hru_cb                          n             n             n             y
hru-lte_wb                      n             n             n             y
...
pest                            n             n             n             y
```

`nyskip` 3 leaves the first three years (2010 to 2012) out of the outputs, so the average annual files cover 2013 to 2023. `basin_wb` writes yearly and average annual files, and every other row writes average annual files only.

An older SWAT+ Editor wrote this file, and 62.0.1 reads some of it differently from its headers:

- On the `csvout dbout cdfout` line, the second value is read as `use_obj_lbls`. It is `n`, so the object rows are read by position, in the fixed order listed on the print.prt page, and their names are not checked.
- On the `soilout mgtout hydcon fdcout` line, the first value is read as `crop_yld`. It is `n`, so no `crop_yld_*` files are written. `mgtout` is `y`, so `mgt_out.txt` is written.
- The fixed order has no `hru_cb` row. The `hru_cb` line is read as the `hru-lte_wb` row, and every later line is read one row early. All these lines have the same switches (`n n n y`), so in this file the shift changes nothing. The carbon rows can only be set by name, with `use_obj_lbls` = `y`.

object.cnt

```
name    ls_area  tot_area  obj  hru  lhru  rtu  mfl  aqu  cha  res  rec  exco  dlr  can  pmp
 ↪  out  lcha  aqu2d  hrd  wro
Osu          10        10    4    1     0    1    0    1    0    0    0     0    0    0    0
 ↪  0     1      0
```

Four objects: one HRU, one routing unit, one aquifer and one SWAT+ channel (`lcha`). The line ends before `hrd` and `wro`, which stay 0. `rec` is 0, so the recall object in `recall.con` is not built (see Other files).

References: time.sim, print.prt, object.cnt.

Basin settings

```
pet_file  wq_file  pet  event  crack  rtu_wq  sed_det  rte_cha  deg_cha  wq_cha  nostress  cn
 ↪  c_fact  carbon  lapse  uhyd  sed_cha  tiledrain  wtable  soil_p  gampt  atmo_dep  stor_max
 ↪  i_fpwet  gw_flow  idc_till
null      null     1    0      0      1       0        0        0        1       0         0
 ↪  0       0       0      1     0        0          0       0       0      a         0
 ↪  0        0        3
```

- `pet` 1: Penman-Monteith potential evapotranspiration, which uses solar radiation, humidity and wind.
- The sixth column has the old header `rtu_wq`, and 62.0.1 reads it as `swift_out`. Its value 1 makes every run write the SWIFT input files into a `SWIFT` folder at the end.
- `wq_cha` 1 is not used by 62.0.1.
- `carbon` 0: static soil carbon, so no carbon file is needed.
- `gampt` 0: curve number runoff.

References: codes.bsn, parameters.bsn.

Climate

`weather-sta.cli` defines one station:

```
name              wgn     pcp            tmp            slr            hmd   wnd
 ↪  wnd_dir  atmo_dep
s35610n127290e    Imsil   Imsilpcp.pcp   Imsiltmp.tmp   Imsilsol.slr   sim   Imsilwind.wnd
 ↪  null     null
```

Precipitation, temperature, solar radiation and wind come from the four Imsil files, each listed in its index file (pcp.cli, tmp.cli, slr.cli, wnd.cli). Relative humidity is generated (sim) from the monthly statistics of the Imsil record in weather-wgn.cli. The precipitation and temperature records run from 1981 to the end of 2023. The solar radiation and wind records end on day 304 of 2022, and the model generates those two variables for the rest of the run (see weather outside the observed record).

References: weather-sta.cli, *.pcp, *.tmp.

HRU, routing unit, aquifer and channel

hru.con has one HRU of 10 ha with no outflow connections (out_tot = 0):

```
id  name     gis_id  area  lat       lon        elev       hru  wst              cst  ovfl
 ↪  rule  out_tot
 1  hru0001       1    10  35.65311  127.36763  350.96629    1  s35610n127290e     0     0
 ↪  0        0
```

The HRU is the only element of routing unit rtu001 (rout_unit.def and rout_unit.ele). The routing unit collects its outflow and passes it on through rout_unit.con: surface, lateral and tile flow (tot) to channel 1, and percolation (rhg) to aquifer 1. The aquifer returns its flow to the same channel through aquifer.con. Channel 1 in chandeg.con has no outflow connection and is the basin outlet. The connection files page shows these three files.

aquifer.aqu has 21 records, aqu011 to aqu102 and aqu_deep010, all with the same parameters except the deep one. Only the first record is used: aquifer.con has one aquifer, and its aqu column points to row 1.

hru-data.hru names the records the HRU uses:

```
id  name      topo          hydro     soil        lu_mgt        soil_plant_init  surf_stor
 ↪  snow     field
 1  hru0001   topohru0001   hyd0001   PadHOEGOG   rice140_lum   soilplant1       paddy0001
 ↪  snow001  null
```

surf_stor = paddy0001 points to a record in wetland.wet whose hydrology record is paddy, which makes the HRU a paddy with ponded water and a weir. The wetland page explains the paddy records in this dataset. A misspelled name in any of the other columns would not be reported: the HRU would use the first record of that file (see name lookup).

References: *.con, hru-data.hru, aquifer.aqu, wetland.wet.

Land use and management

landuse.lum links rice140_lum to the plant community rice140_comm in plant.ini and to the schedule rice140_rot in management.sch, with curve number record rc_strow_g, conservation practice cross_slope and Manning's n record convtill_res. The plant community starts with no growing plant.

rice140_rot has one dated operation and seven decision tables:

```
rice140_rot    1    7
     plow
     weir60r
     weir0
     pad_irr_r
     puddle
     pl_hv_rice
     fert_paddy
manu      5          1      0     beef_salt   broadcast  12.9927
```

The dated operation applies 12.99 kg/ha of the manure beef_salt with broadcast on 1 May each year. The decision tables in lum.dtl do the rest:

Table	Action
plow	Tillage with fieldcultivat early in the year, once the base-zero heat units pass 0.05.
weir60r	Raises the paddy weir to 60 mm on a day between day 120 and day 130.
weir0	Lowers the weir to 0 on a day between day 270 and day 280.
pad_irr_r	From May, while the weir is up and the ponded water is shallower than 60 mm, irrigates with the ponding90 record of irr.ops.
puddle	After April, with the weir up, water ponded and less than 10 mm of rain that day, puddles the paddy with the med_eff record of puddle.ops.
pl_hv_rice	Plants rice140 with the weir up once the base-zero heat units pass 0.15, and harvests and kills it once it has reached 1.15 times its heat units.
fert_paddy	Applies 100 kg/ha of elem_n and 20 kg/ha of elem_p early in the year, and 36 kg/ha of elem_n once the rice has passed half its heat units.

Two shipped files that these tables use are not read as their authors intended:

- The plant action names the transplant record tr_rice140. That record is in transplant.plt, and 62.0.1 reads transplant records only from transplant.ops. The name is not found, no message is written, and the rice is planted from seed. mgt_out.txt still writes a TRANSPLANT line after each PLANT line. With the file renamed to transplant.ops, the model uses the record, and the planting and harvest dates change (see transplant.ops).
- puddle.ops has no title line, so the model reads its header line as the title and its first record, high_eff, as the header. The puddle table names med_eff, which is still read, so puddling works (see puddle.ops).

References: landuse.lum, management.sch, decision tables, *.ops.

Other files

- soils.sol, nutrients.sol, soil_plant.ini: soil PadHOEGOG and its initial nutrients. The soils page and the initial conditions page use Osu_1hru as their examples.
- channel-lte.cha, hyd-sed-lte.cha, initial.cha, nutrients.cha: the channel record used by chandeg.con. See SWAT+ channels.
- recall.con and recall.rec: rec is 0 in object.cnt, so no recall object takes part in the run. The model still reads recall.rec, because file.cio names it, and opens the data file RECYR. named there. That file is not in the folder, so the run leaves an empty file of that name behind. See recall.
- pesticide.pes: the half-life columns contain the old decay rates (0.007 and 0.05), which 62.0.1 reads as half-lives in days. No pesticides are simulated here, so they have no effect (see pesticide.pes).
- salt_hru.ini, salt_aqu.ini, salt_channel.ini: empty, and not used, because no salts are simulated (see salts).
- calibration.cal and cal_parms.cal: the parameter changes described under file.cio, in the format on the parameter change files page.

The shipped folder also contains output files left from runs with older revisions: mgt_out.txt, erosion.txt, fort.2222, the *.swf files at the top level and the SWIFT folder. A 62.0.1 run overwrites mgt_out.txt, erosion.txt and the files in SWIFT, and does not write the others.

Running the model

From inside the folder:

```
path/to/swatplus-62.0.1-gnu-lin_x86_64-Rel
```

With one HRU the run takes a few seconds and writes a few megabytes. The screen shows the date being simulated, and simulation.out ends with:

```
 Original Simulation           12  31  2023 Yr   14 of   14 Time  ...

 Execution successfully completed
```

Besides the output files, the run creates some empty files in the folder: `null`, `RECYR.`, `wtps_wuses.wal`, `cs_hru.ini`, `cs_aqu.ini` and `cs_channel.ini`. They come from readers that create a file when the one they look for is missing, and they do not affect the results.

Outputs

The values below are from a five-year run (2010 to 2014) of a copy of this folder, so the average annual values cover 2013 and 2014 only. A full run gives different numbers.

diagnostics.out

```
 DIAGNOSTICS.OUT FILE
```

No missing names or files are reported.

checker.out

```
sname       hydgrp      zmx   usle_k    sumfc     sumul  usle_p  usle_ls    esco    epco
 ↪  cn3_swf  perco  latq_co  tiledrain
PadHOEGOG   C         750.0   0.0000   98.114   230.336  0.7500   0.1045  0.9500  0.5000
 ↪  0.9500  0.9000   0.0100          0
```

`checker.out` is written before the changes in `calibration.cal` are applied, so `esco`, `epco`, `cn3_swf` and `perco` are the values from `hydrology.hyd`. `usle_k` is 0, so the HRU produces no USLE erosion. The sediment in the outputs comes from puddling, which sets the sediment concentration of the ponded water.

mgt_out.txt

The log of every operation and decision table action. The first season, shortened:

```
 1  2010   2  21  beef_salt   FERT
 1  2010   2  21  beef_salt   FERT
 1  2010   3  24  fldcult     TILLAGE
 1  2010   5   1  beef_salt   MANURE
 1  2010   5   4  paddy       RESET WEIR HEIGHT (m)   0.06
 1  2010   5   4  ponding     PADDY IRRIGATION
 1  2010   5   5  puddle      PUDDLE
 1  2010   5   8  rice140     PLANT
 1  2010   5   8  rice140     TRANSPLANT
 1  2010   7  21  beef_salt   FERT-WET
 1  2010  10   3  paddy       RESET WEIR HEIGHT (m)   0.00
 1  2010  10   8  rice140     HARV/KILL
```

The `FERT` lines from the `fert_paddy` table show the name `beef_salt`, the schedule's dated operation, in place of `elem_n` and `elem_p`. The amounts in the `op_var` columns (100, 20 and 36 kg/ha) are the ones applied. `FERT-WET` means the fertilizer went into the ponded water. Between planting and harvest the log also has a `PLANT_ALREADY_GROWING` line on every day the `pl_hv_rice` table finds the planting conditions met.

The 2010 rice is harvested on 8 October. In the later years of this run it does not reach 1.15 times its heat units before winter and is harvested the following spring, a few days before the next planting. Read this file to check that the decision tables do what was intended.

Water balance

`basin_wb_aa.txt` and `hru_wb_aa.txt` have the same values, because there is one HRU:

precip	irr	surq_gen	perc	et	wet_evap	wet_oflo
1321.9 mm	239.5 mm	830.8 mm	404.5 mm	333.3 mm	116.3 mm	830.1 mm

Most of the rain and irrigation leaves the paddy over the weir (`wet_oflo`), and this water is reported as surface runoff. `basin_wb_yr.txt` gives the same columns for each printed year.

Columns: Water balance outputs.

Routing, aquifer and channel

- `ru_aa.txt` and `hydout_aa.txt`: the routing unit sends about 83,000 m³ a year to the channel and about 40,000 m³ of recharge to the aquifer. The units line of these files says m³/s, but the values are volumes in m³ (see hydrographs).
- `aquifer_aa.txt`: 404.5 mm of recharge and 379.5 mm of return flow to the channel (`flo_cha`).
- `channel_sd_aa.txt`: the channel inflow is the routing unit and aquifer flow together, about 0.004 m³/s on average.

Columns: Aquifer, reservoir, plant, Channel, sediment, nutrient.

Plant growth and the paddy

- `hru_pw_aa.txt`: plant biomass, yield, stresses and weather for the HRU. With `crop_yld` set to `n` in `print.prt` there are no `crop_yld_*` files, so read the yield from this file.
- `wetland_aa.txt`: the ponded water of the paddy, with precipitation, evaporation, seepage and outflow in m³.
- `paddy_test.csv`: a daily table of the paddy (irrigation, seepage, weir height, water depth, sediment and nitrate). 62.0.1 writes it in every run.

Changes to try

- Rename `transplant.plt` to `transplant.ops` and compare the dates in `mgt_out.txt`.
- Set the first value on the `soilout` line of `print.prt` to `b` to get the `crop_yld_*` files.
- Set `pet` to 2 in `codes.bsn` for Hargreaves PET and compare `et` in `hru_wb_aa.txt`.
- Set the sixth value of `codes.bsn` to 0 to stop writing the `SWIFT` folder.
- Shorten the run in `time.sim`, or switch on daily `basin_wb` output, as shown in the cookbook.

Sources

- Dataset: `refdata/Osu_1hru` at 62.0.1.
- Run: a copy of the folder with `time.sim` shortened to 2010 to 2014 and the shipped output files removed, run with a 62.0.1 Linux build (Intel compiler) made from commit 97ca231 on 20 October 2026. That commit differs from the 62.0.1 tag only in gwflow, calibration and export coefficient code, which this dataset does not use. `diagnostics.out`, `checker.out`, `mgt_out.txt` and the output values are from that run. The transplant test used the same copy with `transplant.plt` copied to `transplant.ops`.
- `print.prt` reading: `basin_print_codes_read.f90` reads line 7 into `pco%csvout`, `pco%use_obj_labels`, `pco%cdfout` and line 9 into `pco%crop_yld`, `pco%mgtout`, `pco%hydcon`, `pco%fdcout`. In fixed order, the row after `hru_pw` is `pco%wb_sd` (`hru-lte_wb`), which receives the Osu `hru_cb` line.
- `object.cnt`: `basin_read_objs.f90` reads `bsn, sp_ob` in one list-directed read, and the short line ends the read at end of file.
- `codes.bsn`: `basin_read_cc.f90` reads the line into `bsn_cc`, whose sixth component is `swift_out`. The SWIFT files are written by `swift_output.f90`.
- `checker.out` and calibration: `checker.out` is written in `proc_hru.f90`, and `proc_cal` runs later in `main.f90.in`.
- Transplant lookup: `dtbl_lum_read.f90` matches the `plant` action's file pointer against `transpl%name`, which `plant_transplant_read.f90` fills from `transplant.ops` only. In the `plant` case of `actions.f90`, `mgt_transplant` is called only when `act_app` > 0, but the `TRANSPLANT` line is written whenever `pco%mgtout` = `y`.
- `fertilize` action log: the `fertilize` case of `actions.f90` writes `mgt%op_char` in the name column of `mgt_out.txt`, where `mgt` is the HRU's current schedule operation (`sched%mgt_ops(hru%cur_op)`, set in `mgt_operatn`).

- Recall: recall_read.f90 reads recall.rec whenever the file exists, independent of the rec count, and opens each record's data file without checking that it exists, which creates RECYR..
- Puddle file: mgt_read_puddle.f90 skips a title and a header line before the records.

The SWAT+ source repository has one analysis notebook, jupyter_notebooks/carbon.ipynb. It reads the carbon output of one HRU and draws time series of soil carbon, the carbon pools by soil layer, plant carbon and the carbon fluxes, with the HRU's management operations marked on each graph. It uses the legacy CSU carbon files, which have no interval suffix in their names (see legacy CSU carbon outputs). The newer carbon files, such as hru_cbn_lyr_day.txt, are not read by the notebook.

Files read

The notebook reads these files from one folder:

File	Use
hru-data.hru, landuse.lum	Soil, land use and schedule names of the chosen HRU.
mgt_out.txt	Management operations of the HRU, drawn as markers.
hru_cbn_lyr.txt	Total soil carbon by layer.
hru_seq_lyr.txt	Sequestered soil carbon, total and in the top 300 mm.
hru_plc_stat.txt	Plant carbon: above ground, leaf, stem, seed, root and surface residue.
hru_cpool_stat.txt	Residue, metabolic, structural, lignin, microbial, slow humus and passive humus carbon, root mass and soil water, by layer.
hru_org_trans_vars.txt	Potential transformation rates (bmctp, hsctp, hsntp, hpctp, lslctp, lslnctp) by layer.
hru_cflux_stat.txt	Carbon flows between pools and CO2 losses by layer.

The input files and the output files must be in the same folder, which is the case unless file.cio sets an output folder.

Switches

The legacy files are written only with these settings:

- carbon = 2 in codes.bsn, with carbon.bsn and carbon_lyr.bsn present (see carbon model). hru_plc_stat.txt, hru_cpool_stat.txt, hru_cflux_stat.txt and hru_org_trans_vars.txt are written only with carbon = 2.
- use_obj_lbls = y in print.prt, because the carbon rows are recognised only by name.
- A hru_cb row with y or l in the daily column, for the first five files in the table. l adds layer properties at the start and end of the run.
- A hru_cb_vars row with y or l in the daily column, for hru_org_trans_vars.txt.
- mgtout = y in print.prt, for mgt_out.txt.

The notebook uses the shortest interval in each file, and it is designed for daily output.

Ames_sub1 setup

The Ames_sub1 reference dataset at 62.0.1 already has all of these settings:

```
csvout        use_obj_lbls  cdfout
n             y             n
crop_yld      mgtout        hydcon        fdcout
b             y             n             n
objects                  daily       monthly        yearly         avann
...
hru_cb                     l             y             y             y
hru_cb_vars                l             y             y             y
```

The full run of 46 years writes about 9 GB, most of it daily carbon output. For the notebook, shorten the run first: with `time.sim` set to 1975 to 1976, the run takes a few seconds and the output folder is about 380 MB.

Running the notebook

1. Copy `refdata/Ames_sub1` to a new folder, shorten `time.sim`, and run SWAT+ in the copy (see Running the model).
2. Install the Python packages. The notebook uses `pandas` and `plotly`, and plotly needs `nbformat` to show figures in Jupyter:

   ```
   pip install jupyterlab pandas plotly nbformat
   ```

3. Open `carbon.ipynb` in JupyterLab or VS Code.
4. In the first code cell, under `# hru to run against.`, set:
   - `hru_id`: the HRU number to plot (1 by default).
   - `wdir` and `sdir`: the notebook reads from the folder `f"{wdir}/{sdir}"`. The values in the shipped notebook are the author's folders and must be changed, for example `wdir = "/home/me/runs"` and `sdir = "Ames_sub1"`.
   - `output_html_file`: `True` also saves every figure as an HTML file.
5. Run all cells.

The cell that reads a file does nothing when the file is missing, and the graph cells after it then work with the table of the previous file, or stop with an error if there is none. Turn on every switch listed above before running the notebook.

Graphs

Each section of the notebook has a cell that reads one file and one or more cells that draw Plotly figures from it. With daily output for HRU 1 of Ames_sub1, the notebook draws, in order:

1. Total soil carbon by layer (`hru_cbn_lyr.txt`), in Mg/ha per mm of layer thickness.
2. Sequestered carbon in the top 300 mm (`hru_seq_lyr.txt`), in Mg/ha.
3. Plant carbon components (`hru_plc_stat.txt`), in kg/ha.
4. Residue and soil carbon pools in the 10 mm top layer, and in the layer at a depth set in the cell (100 mm by default) (`hru_cpool_stat.txt`).
5. One graph by layer for each pool in `hru_cpool_stat.txt`: residue, metabolic, structural, lignin, microbial, slow humus, passive humus, live root mass and soil water content.
6. One graph by layer for each potential transformation rate in `hru_org_trans_vars.txt`.
7. One graph by layer for each carbon flow between pools (`cfmets1` to `cfs3s1`) and each CO2 loss (`co2fmet` to `co2fs3`) in `hru_cflux_stat.txt`.

Several graph cells have a `normalize` setting that divides each layer's value by the layer thickness, and a `show_mgt` setting that turns the management markers on or off. The markers are coloured by operation type: planting, harvest, tillage, fertilizer, manure, irrigation, pesticide and grazing.

Related pages

- Carbon outputs: the newer carbon files and the legacy files with their columns.
- carbon.bsn: carbon model parameters and initial pools.
- Soil carbon theory: what the pools and flows represent.
- Cookbook: carbon model: switching the carbon model on in another project.

Sources

- Notebook: `jupyter_notebooks/carbon.ipynb` at 62.0.1. The file names, the folder variables `wdir` and `sdir`, the `read_data` and `read_mgt_out` functions and the graph cells are in its code cells.

- Test: a copy of `refdata/Ames_sub1` with `time.sim` set to 1975 to 1976, run with a 62.0.1 Linux build (Intel compiler) made from commit 97ca231, which differs from the tag only in gwflow, calibration and export coefficient code. The run took about 10 seconds and wrote 379 MB. All code cells of the notebook were then run outside Jupyter with pandas 3.0.1 and plotly 6.6.0, with `wdir` pointed at the copy; every cell completed and 33 figures were drawn, with 731 daily values (1975 and 1976) in each data trace.
- Legacy file switches: `command.f90` calls `soil_nutcarb_write_legacy.f90` for `pco%cb_hru` and `soil_carbvar_write_legacy.f90` for `pco%cb_vars_hru` (only with `bsn_cc%cswat` = 2). The files are opened in `carbon_legacy_module.f90`. The `freq` column takes the values `day`, `mon` and `year`, which the notebook's reading cells filter on.
- Missing files: each reading cell tests `file_path.is_file()` and otherwise only sets `df_plot` to an empty table; the graph cells test `df.empty`, where `df` is the table from the last reading cell that found its file.

Six common edits to a SWAT+ project that already runs. Each recipe names the file and the values to change, and what to check after the run. The input reference pages linked from each recipe give the full rules.

Simulation period

Edit `time.sim`:

```
day_start  yrc_start   day_end   yrc_end      step
        0       1975         0      2020         0
```

- `yrc_start` and `yrc_end` are the first and last calendar years.
- `day_start` is the day of the year the run starts; 0 is read as 1, 1 January.
- `day_end` is the day of the year the run ends in the last year; 0 means 31 December.

To leave warm-up years out of the outputs, keep `time.sim` and set `nyskip` on line 3 of `print.prt` to the number of years to skip. The skipped years are still simulated.

Days outside the measured weather records are generated from `weather-wgn.cli` without a message, so a period longer than the records still runs. Compare the period with the first and last dates in the `.pcp` and `.tmp` files before reading the results.

References: time.sim, print.prt, weather outside the observed record.

Daily or monthly output

Each object row of `print.prt` has four switches, for daily, monthly, yearly and average annual files. Set the column to `y` to write that file, for example daily basin water balance:

```
objects                 daily       monthly        yearly         avann
basin_wb                    y             n             y             y
```

How the rows are read depends on the second value on line 7 (`use_obj_lbls`, labelled `dbout` in files from older Editors):

- `y`: rows are matched by name. A row can be added, removed or moved, and the carbon and gwflow rows can be used.
- `n`: rows are read by position in a fixed order. Do not add, remove or reorder rows: every later row would apply to the wrong output, without a message.

Daily output of many objects over a long period can take several gigabytes, so check the free disk space before a long run. After the run, `files_out.out` lists every output file that was opened.

References: print.prt, Output naming convention.

PET method

Edit `pet`, the third value in codes.bsn:

pet	Method	Weather it uses
0	Priestley-Taylor	temperature and solar radiation
1	Penman-Monteith	temperature, solar radiation, relative humidity and wind
2	Hargreaves	temperature
3	weather station PET	in 62.0.1 this gives Hargreaves PET (see below)

The PET method does not change which weather files are read. A variable that the station takes from the weather generator (`sim` in weather-sta.cli) is generated in every case, so Penman-Monteith with generated humidity and wind uses generated values. With `pet` = 3, 62.0.1 replaces the measured station PET with Hargreaves PET before the HRUs use it, so this code gives Hargreaves PET times `pet_co`.

After the run, compare `pet` and `et` in `hru_wb_aa.txt` or `basin_wb_aa.txt` with the earlier run.

References: PET method, weather-sta.cli.

Green-Ampt infiltration

Three files change:

1. codes.bsn: set `gampt`, the 21st value, to 1.
2. time.sim: set `step` to the number of time steps per day, for example 24 for hourly steps.
3. The precipitation files: give sub-daily records with the same number of values per day as `step`. With `step` above 1, 62.0.1 does not split a measured daily record over the time steps, so a daily `.pcp` file gives no rain in any time step.

Plant growth, soil water and nutrient cycling still run once a day.

After the run, compare `surq_gen` and `perc` in `basin_wb_aa.txt` with the curve number run.

References: runoff method, time step, *.pcp.

HRU addition

An HRU needs a row in several files. Copy the rows of an existing HRU and change them:

1. hru.con: add a row with the next `id`. Rows must be numbered 1, 2, 3 and so on in file order. Set `area`, `lat`, `elev`, the `hru` column (the row number of the new HRU in hru-data.hru) and `wst` (a station name in weather-sta.cli).
2. hru-data.hru: add a row whose `topo`, `hydro`, `soil`, `lu_mgt`, `soil_plant_init`, `snow` and `field` names exist in the corresponding files. A misspelled name is not reported: the HRU uses the first record of that file.
3. The HRU's outflow: either give it outflow connections in hru.con (`out_tot` and the groups after it), or add it as an element of a routing unit in rout_unit.ele and the element list in rout_unit.def, as in Osu_1hru. With neither, its water leaves the model.
4. ls_unit.ele and ls_unit.def, if the project has them: add the HRU as an element. The basin output files (`basin_wb` and the others) are added up from these elements with their `bsn_frac` values as given, so the `bsn_frac` values should add up to 1 after the change.
5. object.cnt: add 1 to `hru` and to `obj`.

After the run, `checker.out` has one row per HRU, in HRU order, with the soil and hydrology values the new HRU ended up with. Check `diagnostics.out` for names that were not found, and remember that the `topo`, `hydro`, `soil`, `lu_mgt`, `soil_plant_init` and `snow` names are not checked there.

References: *.con, hru-data.hru, object.cnt, print.prt.

Carbon model

The dynamic soil carbon model needs three changes:

1. `codes.bsn`: set `carbon`, the 14th value, to 2. The value 1 is reserved for C-FARM, which is not implemented, and runs the model with no organic matter cycling at all. Datasets from before revision 62 used 1 for the dynamic model and must be changed.
2. `file.cio`: name the carbon file as the third name on the `basin` line, for example `basin codes.bsn parameters.bsn carbon.bsn`.
3. Provide `carbon.bsn` and `carbon_lyr.bsn` in the project folder. `refdata/Ames_sub1` has both, and they can be copied as a start. SWAT+ stops with an error if either is missing. The old `carb_coefs.cbn` is no longer read.

For carbon output, `print.prt` must use `use_obj_lbls = y`, and the carbon rows are added by name. The newer rows (`hru_cb_gl`, `hru_cb_lyr`, `hru_cb_cpool` and others) each switch one family of files, and the legacy rows `hru_cb` and `hru_cb_vars` switch the files the carbon notebook reads. Daily per-layer carbon output is large: in Ames_sub1 it makes up most of the 9 GB of a full run, so start with a short period.

After the run, `files_out.out` lists the carbon files that were opened, and Carbon outputs describes their columns.

References: carbon model, carbon.bsn, older file.cio files, carb_coefs.cbn.

Sources

- Simulation period: `time_read.f90` sets `time%day_start` to 1 when it is 0 or less.
- Output rows: `basin_print_codes_read.f90` reads the second value of line 7 into `pco%use_obj_labels` and the object rows by position or by name accordingly.
- `codes.bsn`: `basin_read_cc.f90`. Column positions: `pet` 3 (`bsn_cc%pet`), `carbon` 14 (`bsn_cc%cswat`), `gampt` 21 (`bsn_cc%gampt`). PET with `pet = 3` and sub-daily precipitation: see the Sources sections of codes.bsn and time.sim.
- HRUs: connection rows are read by `hyd_read_connect.f90`, HRU rows and their name lookups by `hru_read.f90`, counts by `basin_read_objs.f90`, and `checker.out` is written by `proc_hru.f90`. Basin output weighting by `bsn_frac`: see nutrient balance and losses.
- Carbon: the `basin` line of `file.cio` is read in `readcio_read.f90`, and `carbon.bsn` and `carbon_lyr.bsn` by `carbon_bsn_read.f90`.

The SWAT+ Fortran source has a generated API reference, built with FORD and published at

swat-model.github.io/swatplus

It is rebuilt for every release tag; on 20 October 2026 it shows "SWAT+ Model Sources (62.0.1)". The reference contains:

- every module, derived type, subroutine and function in `src/`;
- call and caller graphs;
- the source of each `.f90` file;
- the comment block at the top of each routine, searchable.

This site and the FORD reference

Use this site for input and output files, model theory, calibration and running the model. Use the FORD reference to find what a routine does, which variables it reads and writes, and where it is called from. The collapsed Sources block at the end of each page names the routines behind it, which are the starting points in FORD.

Local build

The FORD settings are in `ford.md.in` in the source repository; CMake fills in the version and writes `ford.md`:

```
cmake -B build -D TAG=$(git describe --tags --abbrev=0)
ford ford.md
```

The HTML is written to `build/doc`. FORD needs Graphviz for the call graphs.

Sources

- `ford.md.in` and `.github/workflows/doc.yml` (runs on tags matching `*.*` and `*.*.*`, builds with `ford ford.md` and publishes `build/doc` to GitHub Pages).